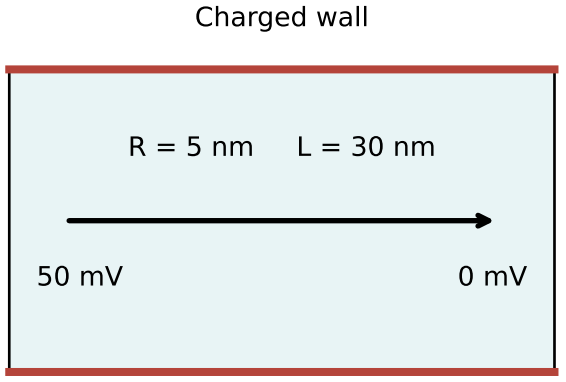
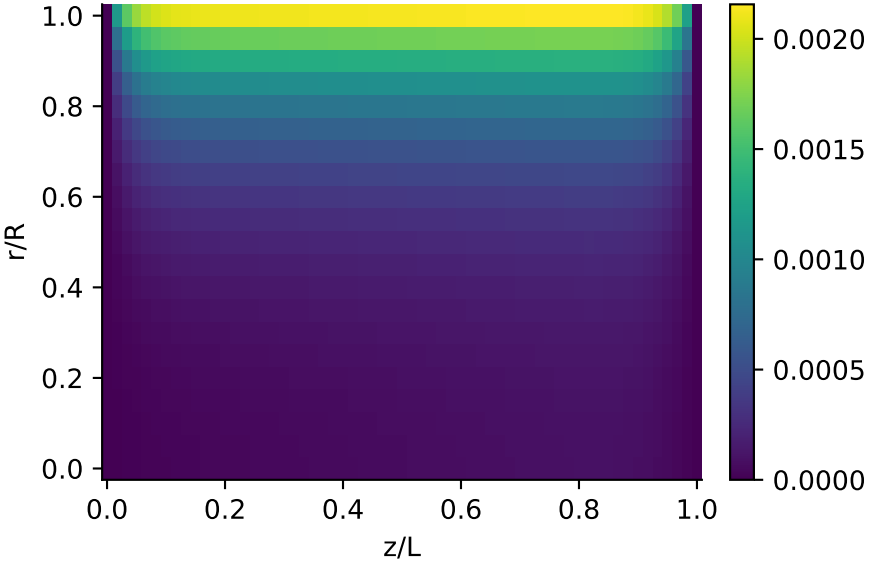
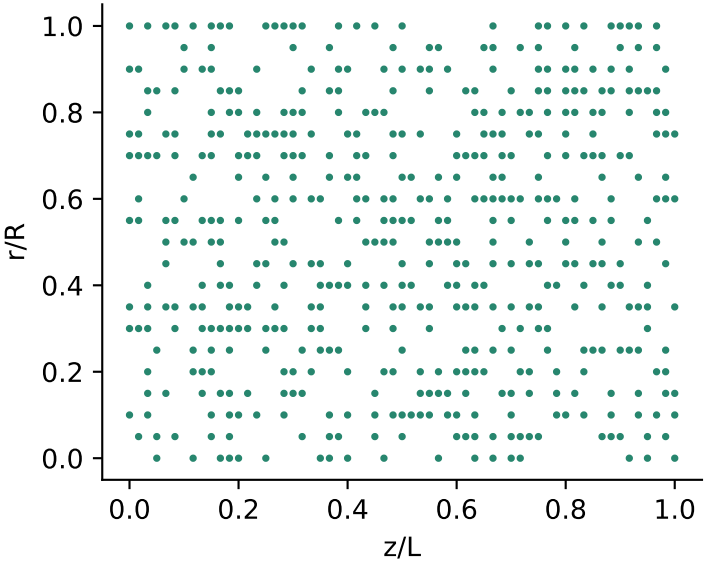
A Cylindrical pore



B Conservative reference: cation excess



C 500 numerical anchors



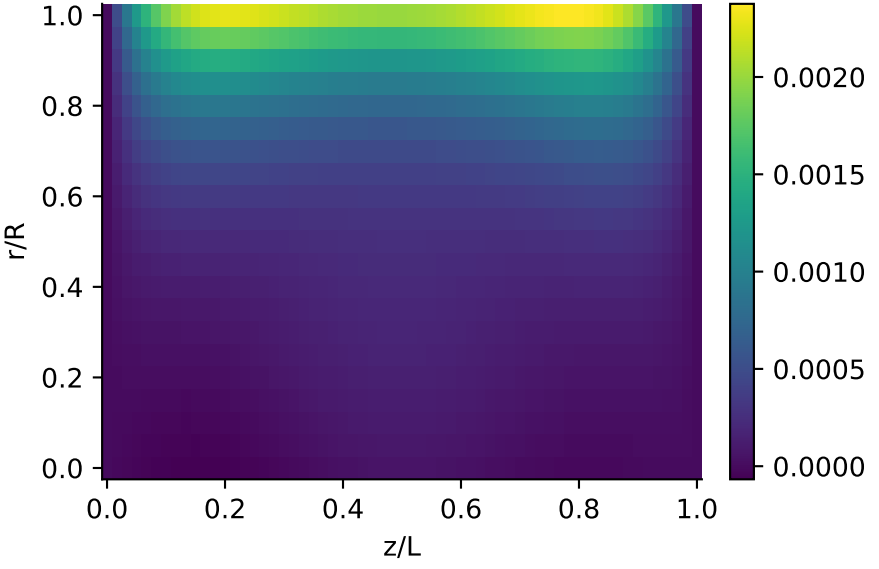
D Exact reservoir constraints

$$\psi_\theta = \psi_{app}(1-z) + z(1-z)N_\psi$$

$$c_{\pm,\theta} = \exp[z(1-z)N_\pm]$$

PNP residuals + axis/wall conditions
+ field and charge assimilation

E DA-PINN: cation excess



F Independent diagnostics

Equilibrium and Ohmic tests

Tolerance and field refinement

Held-out nodes and paired seeds

Charge and current errors

Training and inference timing